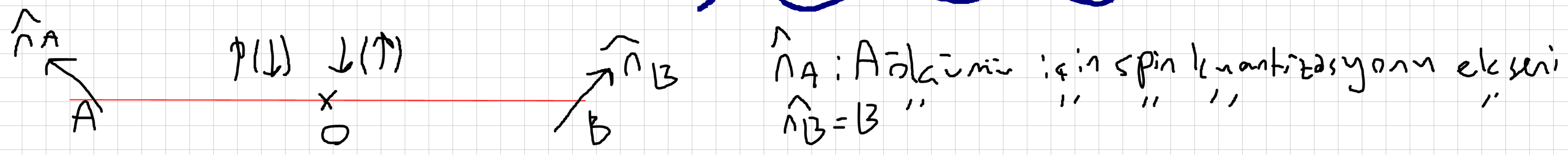


# Quantum Bilim Ders 7 Bell Eritisi Eklili



$$\frac{1}{\sqrt{2}} (|\frac{1}{2}, -\frac{1}{2}\rangle - |-\frac{1}{2}, \frac{1}{2}\rangle)$$

Deney I  $\hat{A}, \hat{B}$  anty paralel

|   | ölçüm 1 | ölçüm 2 | ölçüm 3 | ... |
|---|---------|---------|---------|-----|
| A | +       | -       | +       | -   |
| B | +       | -       | +       | -   |

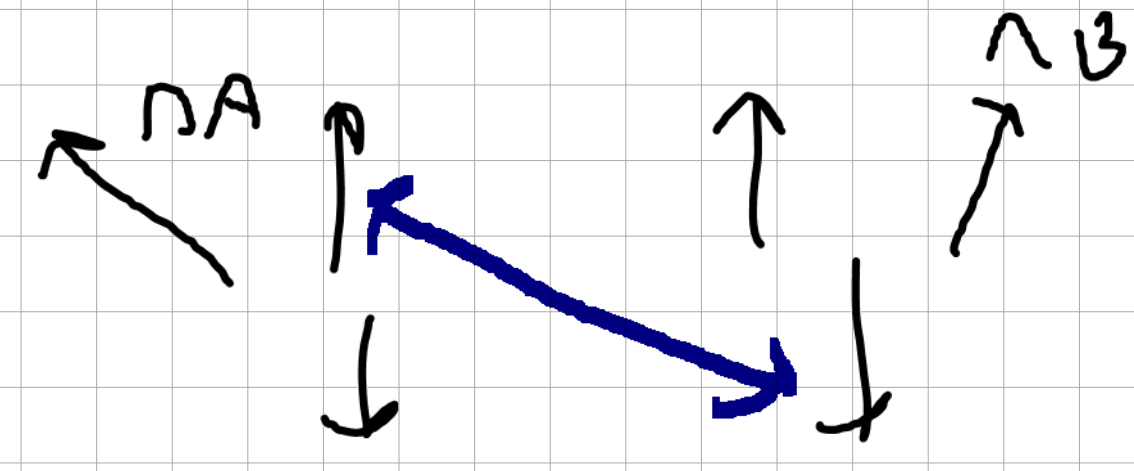
antyparalellik 2 si çözü yada tersi

$$C = \frac{(1)(1) + (-1)(-1) + (1)(1) + \dots}{N} = 1$$

Deney 1  
o/o 100  
Korelasyon

Deney 2  $\hat{A}$  ve  $\hat{B}$  paralel olsun.

|   | ölçüm 1 | ölçüm 2 | ölçüm 3 | ... |
|---|---------|---------|---------|-----|
| A | +       | -       | +       | -   |
| B | -       | +       | -       | +   |



2 si aynı anda ne (+) ne de (-) olur.

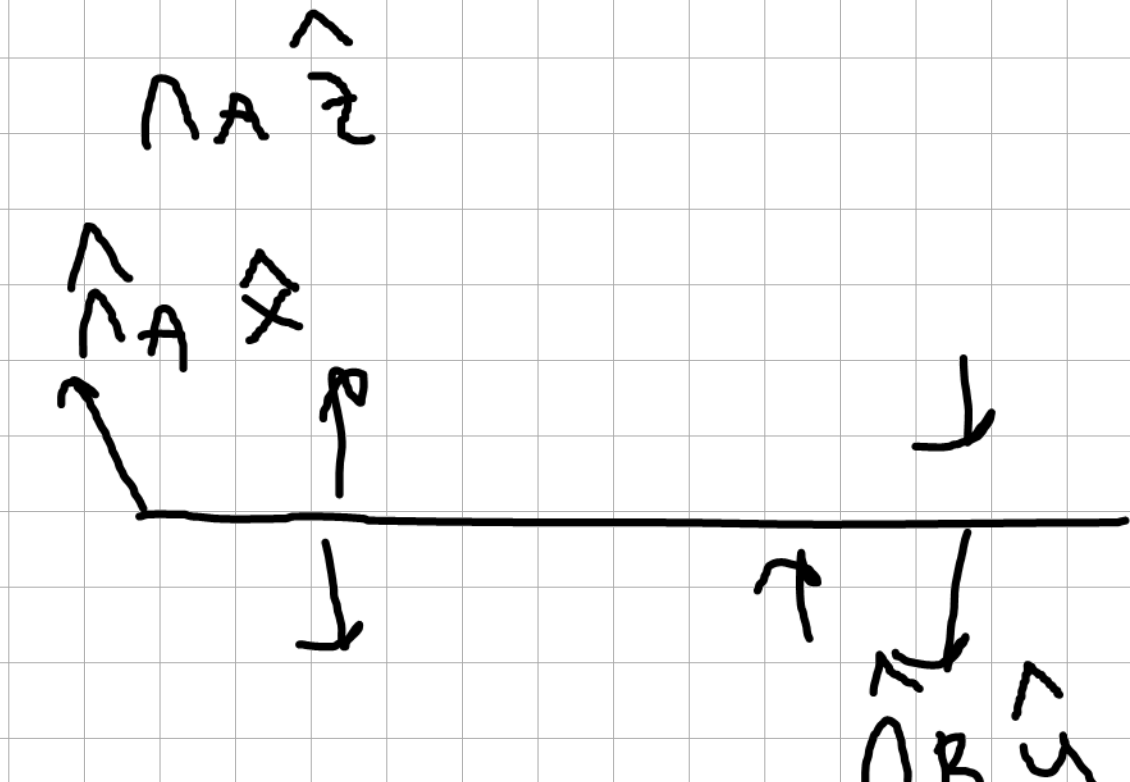
$$C = \frac{(+1)(-1) + (-1)(+1) + (+1)(-1) + \dots}{N} = -1$$

Deney 2 de o/o 100 Antikorelasyon

Deney 3 de  $\hat{A} \perp \hat{B}$

Deney III

|   | ölçüm 1 | ölçüm 2 | ölçüm 3 | ... |
|---|---------|---------|---------|-----|
| A | +       | -       | +       | -   |
| B | -       | +       | +       | -   |



$$|\uparrow_z\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\downarrow_z\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|\uparrow_x\rangle = \frac{1}{\sqrt{2}} (|\uparrow_z\rangle + |\downarrow_z\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|\uparrow_y\rangle = \frac{1}{\sqrt{2}} (|\uparrow_z\rangle - i|\downarrow_z\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$C = \frac{(+1)(-1) + (-1)(+1) + (+1)(+1) + (-1)(-1) + \dots}{N}$$

$$C = \frac{-1 + -1 + 1 + 1 + \dots}{N} = 0$$

$C = 0/00$  Korelasyonsuz

$$\hat{A} = \hat{z} \quad \hat{B} = \hat{x}$$

B de x ölçünce, A da o/050 olur

$$|\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}} (|\frac{1}{2}, -\frac{1}{2}\rangle - |-\frac{1}{2}, \frac{1}{2}\rangle)$$

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow, \downarrow\rangle - |\downarrow, \uparrow\rangle) = \frac{1}{\sqrt{2}} (|+, -\rangle - |-, +\rangle)$$

Deney 1  $\hat{A} = \hat{z}$   $\hat{B} = -\hat{z}$  antiparalel o/o 100  $C_R = 1$

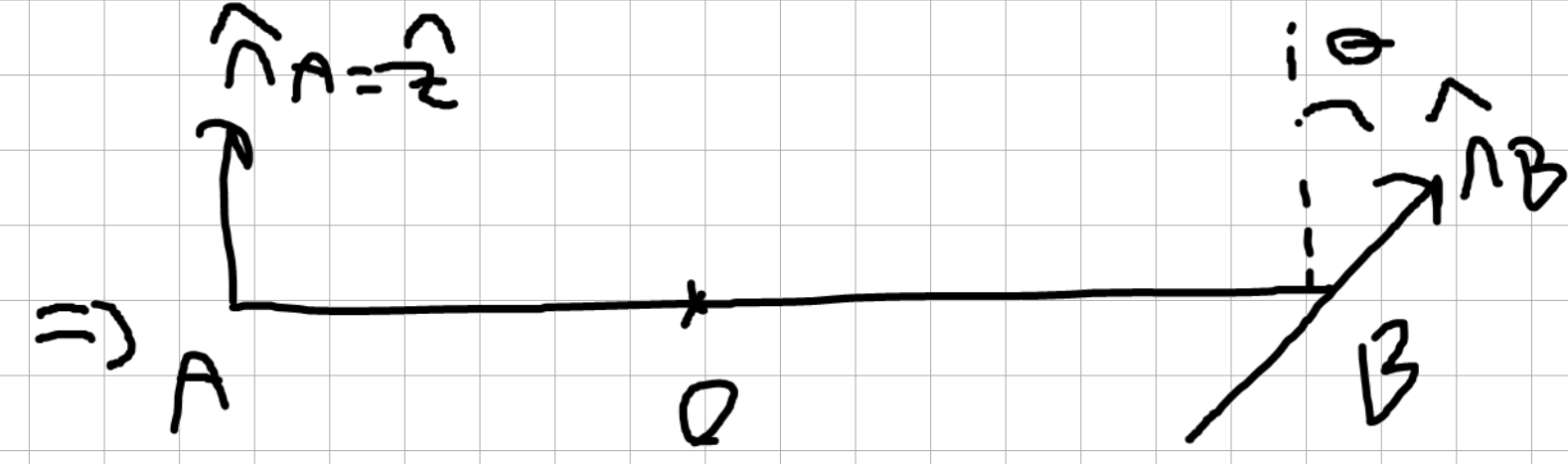
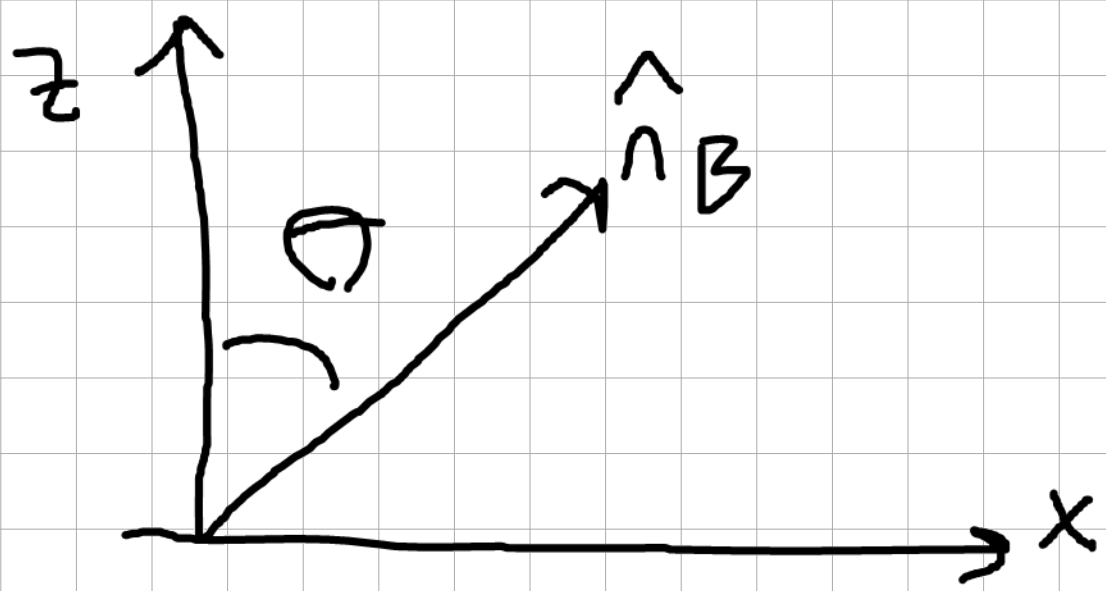
Deney 2  $\hat{A} = \hat{z}$   $\hat{B} = \hat{z}$  paralel o/o 100  $C_R = -1$   
antikorelasyon

Deney 3  $\hat{A} = \hat{z}$   $\hat{B} = \hat{x}$  dik o/o 00  $C_R = 0$   
korelasyonsuz

Deney 4  $\hat{A} = \hat{z}$   $\hat{B} = \cos\theta \hat{z} + \sin\theta \hat{x}$



$$\hat{n}_A = \hat{z} \quad \hat{n}_B = \cos\theta \hat{z} + \sin\theta \hat{x}$$



$P(+,+) = A$  ve  $B$  de  $\hat{z}$  kütlen spinlerin  $+1$  ve  $+1$  çıkma olasılığı

$$P(+,+) \quad P(+,-) \quad P(-,+) \quad P(-,-)$$

$$C_Q = \frac{P(+,+)N + P(+,-)N + P(-,+)N + P(-,-)N}{N} \Rightarrow C_Q = P(+,+) + P(+,-) + P(-,+) + P(-,-)$$